

Math Bootcamp

Day 3 — Remaining real analysis and linear algebra

Math Camp 2026

How to use these slides

Main deck: definitions, recipes, examples, and practice.

Appendix and proofs (marked on relevant slides): **for interest only** — optional recap for those already comfortable with the math.

Not required for Math Camp. Focus on the main slides and exercises unless you want the extra detail.

3.1 Fixed points — equilibrium as a solution

Fixed point. A point x^* is a **fixed point** of a map T if $T(x^*) = x^*$.

Economics reading. Many equilibria are fixed-point problems:

- **Market clearing:** a price p^* with excess demand $E(p^*) = 0$.
- **Game theory:** a strategy profile that is a best response to itself (Nash).

Two useful existence tools (today).

- **Contraction Mapping Theorem:** existence and uniqueness of a fixed point; iteration converges.
- **Brouwer:** continuous map on a closed interval / convex set.

3.1 Contraction Mapping Theorem

Contraction. A map T **shrinks distances** if there is $k \in (0, 1)$ such that $|T(x) - T(y)| \leq k|x - y|$ for all x, y in the domain.

Contraction Mapping Theorem. If $T: [a, b] \rightarrow [a, b]$ is a contraction, then

- **Existence:** there is a fixed point x^* with $T(x^*) = x^*$.
- **Uniqueness:** that fixed point is **unique**.
- **Iteration:** for any starting value $x_0 \in [a, b]$, the sequence $x_n = T(x_{n-1})$ converges to x^* .

3.1 Why iteration converges

Step 1 — Start anywhere. Pick any x_0 . Define $x_n = T(x_{n-1})$ for $n = 1, 2, \dots$

Step 2 — Jumps shrink exponentially. Each new step is at most k times the previous jump:

$$|x_{n+1} - x_n| = |T(x_n) - T(x_{n-1})| \leq k|x_n - x_{n-1}| \leq k^n|x_1 - x_0|.$$

Since $k < 1$, the moves get **exponentially smaller** — the path barely changes after a while.

Step 3 — The sequence converges. Step 2 implies $x_n \rightarrow x^*$ for some limit x^* (check it is a **Cauchy sequence** for a rigorous proof). So iteration from **any** starting point drives x_n toward the same x^* .

Step 4 — The limit is a fixed point. T is continuous, so taking limits in $x_n = T(x_{n-1})$ gives $x^* = \lim x_n = \lim T(x_{n-1}) = T(\lim x_{n-1}) = T(x^*)$.

3.1 Walrasian price adjustment

Market clearing. A price p^* **clears the market** when excess demand is zero: $E(p^*) = 0$. Here $E(p)$ can take **any form** (from supply and demand behind the scenes).

Price adjustment (nonlinear in E). Update the price each period:

$$p_{t+1} = T(p_t) = p_t + \lambda(1 - e^{-\alpha E(p_t)}), \quad \lambda = \frac{1}{5}, \alpha = 0.1.$$

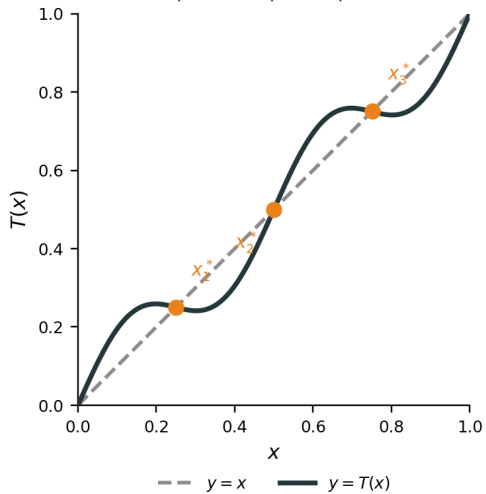
When $|E|$ is small, $1 - e^{-\alpha E} \approx \alpha E$ — move a fraction of excess demand toward balance. Clearing satisfies $E(p^*) = 0 \Leftrightarrow T(p^*) = p^*$. If T is a **contraction** on a price range, then $p_t \rightarrow p^*$ from any starting p_0 .

3.1 Brouwer — one dimension

Brouwer (one dimension). If $T: [a, b] \rightarrow [a, b]$ is **continuous**, then T has **at least one** fixed point x^* with $T(x^*) = x^*$.

Picture. Plot $y = T(x)$ against $y = x$. Any crossing is a fixed point. Continuity forces the graph to meet the diagonal at least once on $[a, b]$.

Multiple fixed points possible



3.1 Brouwer — general statement

Brouwer (several variables). Let $K \subset \mathbb{R}^n$ be **nonempty**, **compact**, and **convex**. If $T: K \rightarrow K$ is **continuous**, then T has at least one fixed point $x^* \in K$ with $T(x^*) = x^*$.

Camp vocabulary.

- **Compact** in $\mathbb{R}^n$: closed and bounded (Heine–Borel).
- **Convex**: the segment between any two points in K stays in K .

One dimension. $K = [a, b]$ is compact and convex, so the 1D case is a special case.

3.1 Brouwer — Cournot duopoly

Two firms. Firm i chooses $q_i \in [0, \bar{q}]$. **Nonlinear** inverse demand $p = a - (q_1 + q_2)^2$ (e.g. $a = 8$, $\bar{q} = 2$).

Profit and FOC. $\pi_i = (a - (q_1 + q_2)^2)q_i$. Firm 1's best response solves

$$a - 3q_1^2 - 4q_1q_2 - q_2^2 = 0 \quad (\text{similarly for firm 2 with } q_1, q_2 \text{ swapped}).$$

The map $q_j \mapsto BR_i(q_j)$ is **nonlinear** — no formula like “ $(a - c - q_j)/2$ ”.

Fixed point = Nash. $T(q_1, q_2) = (BR_1(q_2), BR_2(q_1))$ is continuous on $K = [0, \bar{q}]^2$. **Brouwer** guarantees an equilibrium, but you generally solve $T(q_1^*, q_2^*) = (q_1^*, q_2^*)$ by **algebra or numerics** (here one symmetric solution is $q_1^* = q_2^* = \sqrt{a/8}$).

3.2 Separating Hyperplane Theorem

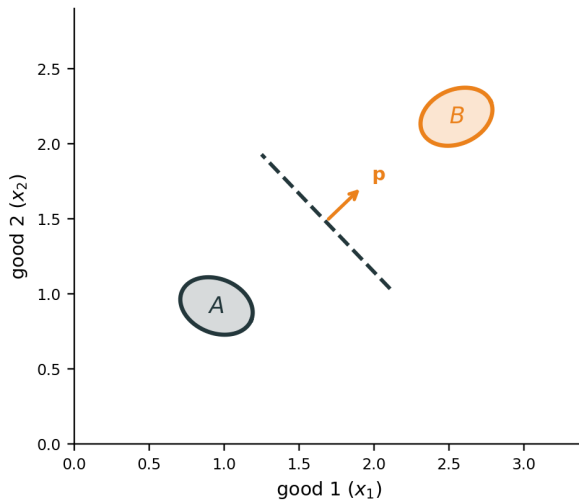
Setup. Two nonempty **convex** sets $A, B \subset \mathbb{R}^n$ with $A \cap B = \emptyset$.

Theorem. If A is **compact**, then there exist $\mathbf{p} \neq \mathbf{0}$ and $\beta \in \mathbb{R}$ such that $\mathbf{p} \cdot a \leq \beta \leq \mathbf{p} \cdot b$ for all $a \in A$, $b \in B$. Geometrically: the hyperplane $\mathbf{p} \cdot x = \beta$ separates A from B .

In two goods ($x = (x_1, x_2)$). The line $p_1x_1 + p_2x_2 = \beta$ puts all of A on one side ($\mathbf{p} \cdot a \leq \beta$) and all of B on the other ($\mathbf{p} \cdot b \geq \beta$); $\mathbf{p}$ is the normal, β fixes the line's position.

Remark (single point). If $B = \{b_0\}$ is one point outside a closed convex set A , a separating hyperplane exists even when A is not compact.

Disjoint convex sets separated by a line



3.2 Example — production set vs. preferred bundles

Classic micro example. Two goods. Let

- A = **production possibility set** (feasible output combinations);
- B = bundles **strictly preferred** to the status quo $\bar{x}$ (better indifference curves).

If preferences are convex, B is convex. If no feasible bundle is strictly preferred to $\bar{x}$, then $A \cap B = \emptyset$.

Separation $\Rightarrow$ prices. The theorem gives $\mathbf{p}^*$ and β with $\mathbf{p}^* \cdot a \leq \beta \leq \mathbf{p}^* \cdot b$ for all $a \in A$, $b \in B$. At a boundary point $a^* \in A$, the line $\mathbf{p}^* \cdot x = \beta$ is a **supporting price line**: the firm produces a^* where profit $\mathbf{p}^* \cdot a$ is maximized over A .

Numbers. If $\mathbf{p}^* = (2, 5)$ and $\beta = 100$, then $a = (10, 16)$ lies on the line ($2 \cdot 10 + 5 \cdot 16 = 100$) while a preferred bundle $b = (14, 18)$ costs $118 > 100$ — affordable production stays on one side, strictly better bundles on the other.